

An experimental investigation of conjunct agreement in Turkish

Özge Bakay and Faruk Akkuş
University of Massachusetts Amherst

ABSTRACT

In several languages when there is agreement with a conjoined phrase, the Agree head can have the sum of the features of all (i.e., *full agreement*) or some conjuncts only (i.e., *partial agreement*). Investigations into (partial) agreement with conjoined phrases have played a crucial role in linguistic theorizing of agreement. Here we aim to contribute to this literature by empirically examining verbal agreement strategies in Turkish. Specifically, we ask to what extent partial agreement on the verb is available when the subject is a coordination or disjunction phrase. Results from three acceptability judgment tasks show for coordination phrases that while Turkish speakers have a preference for full agreement, partial agreement with the last (or second) conjunct ‘I’ is available with both preverbal or postverbal subjects to a certain degree. For preverbal disjunction phrases, we provide novel evidence that the acceptability of both full and partial agreement is overall degraded, suggesting evidence for a grammatical gap in assigning verb number with disjunction. Nevertheless, full agreement is the most available strategy, followed by partial agreement with the last conjunct.¹

1 Introduction

It is widely observed and empirically attested across several languages that the target of agreement can be only *one* or more of the conjuncts within a conjoined phrase (e.g., Koutsoudas 1968). That is, the agreement head may bear the features (e.g., ϕ , noun class) of only the second (as in 1a) or only the first conjunct (as in 1b), resulting in partial agreement. It is also possible for two agreement heads to bear the features of two different conjuncts (as in 1c). Partial agreement is often available alongside full (or resolved) agreement (i.e., agreement with the sum of the features of both conjuncts, as in 1a) or default agreement (i.e., morphological last resort when no feature value is provided) (see Nevins & Weissner 2019, for a review). The examples in 1a-1c illustrate partial or full agreement (features bolded) in Turkish, Moroccan Arabic, and Polish, respectively.

- (1) a. [Sen-in-le ben-im] ara-m/mız-da kocaman bir
 you-GEN-COM **I-GEN** between-**1SG/1PL.POSS-LOC** huge a
 fark var.
 difference exist
 ‘There is a huge difference between you and me.’

¹We would like to thank Brian Dillon, Burak Öney, Kyle Johnson, John Kingston, Utku Türk, Duygu Demiray, Adrian Staub, Suet-Ying Lam, the Psycholinguistics and Syntax Workshops at UMass Amherst and the audience at Tu+10 for their valuable comments. Authors: Özge Bakay (obakay@umass.edu) & Faruk Akkuş (fakkus@umass.edu).

- b. mšite [ntuma w ana] mǝžmuʕin
 left.2PL you.2PL and I together
 ‘You and I left together.’
- c. Maria chce, zebym [ja i mój sąsiad] wyszed
 Maria wants that.COND-1SG I and my neighbor.M.SG left.M.SG
 ‘Maria wants me and my neighbor to leave.’

Structurally, coordinated phrases are often represented by a coordination head (&), whose specifier and complement positions are asymmetrically filled by the first and second conjuncts, respectively (e.g., Benmamoun *et al.* 2009, Munn 1993, cf. Haskell & MacDonald 2005). This is shown in 2.



A common theoretical assumption is that neither the coordination head (&) nor its maximal projection (&P) carries ϕ features (e.g., Citko 2018). Consequently, when an Agree head targets a coordinated phrase, its features cannot be valued by the phrase itself, potentially leading to non-standard agreement routes. Given the asymmetric structure in 2, partial agreement with the first conjunct, as in Example 1b, can be explained by the hierarchically prominent position of the first conjunct. However, partial agreement with the last, linearly closer conjunct of a preverbal phrase, as illustrated in Turkish (Example 1a), is particularly notable for syntactic theories of agreement. This is because agreement with the second conjunct requires the Agree head to access the second conjunct by virtue of its linear proximity rather than its hierarchical position or c-command relations. This challenge is heightened by the fact that languages exhibiting sensitivity to linear proximity often do not show the same sensitivity in other grammatical contexts.

Existing theoretical approaches offer contrasting views on whether linearity-sensitive partial agreement should be analyzed within the syntax proper or assigned to a post-syntactic component. One group of accounts derives partial agreement entirely in the syntax. Mechanisms that enable partial agreement in these accounts include the equidistant positioning of the coordination phrase (&P) and the conjunct in its specifier relative to the Agree head (as argued for Dutch by van Koppen 2005), the presence or absence of movement operations in the language or the ordering of Agree and Merge operations (as proposed for Serbo-Croatian by Bošković 2009 and Murphy & Puškar 2018, respectively). However, a major limitation of these accounts is their lack of generalizability; the language-specific structural and/or operational mechanisms that they rely on often limit their ability to explain partial agreement cases across different languages.

Another group of accounts adopts a two-stage agreement process that incorporates a post-syntactic component (e.g., Benmamoun *et al.* 2009, Bhatt & Walkow 2013, Köylü 2018, Marušič *et al.* 2015, Tat & Kornfilt 2022). In this line of research, after a relation is established between an Agree head and its target in the

syntax, the features of one or more conjuncts are transferred to the head post-syntactically, (i.e., at the PF). Crucially, this second step may be sensitive to linear proximity such that the features of the conjunct that is linearly closer to the head may sometimes be copied, resulting in partial agreement.

Compared to the relatively extensive literature on agreement with coordination phrases, previous work on agreement with disjunction phrases (e.g., those using operators like ‘or’) is limited. Although coordination and disjunction phrases are often considered syntactically similar and both are assumed to lack inherent grammatical number, they differ semantically: Coordination phrases typically denote semantic plurality (unless they refer to a single entity), whereas disjunction phrases do not (e.g., Peterson 2004). This semantic difference may lead to a scenario where speakers’ decisions about verb number with disjunction phrases are often ad hoc and non-deterministic due to the absence of a clear grammatical principle, creating what is sometimes termed *grammatical gap* (Foppolo & Staub 2020).

This view is supported by empirical findings. When the subject is a disjunction of two singular nouns, both English and Italian speakers were found to have degraded ratings for both singular and plural verb agreement, suggesting that neither form is fully acceptable (Foppolo & Staub 2020). Furthermore, English speakers had comparable acceptability and showed no evidence for disruption in eye-movements when encountering verbs with either form, providing further support that both forms may be available to a similar extent. As in conjunction phrases, agreement with disjunction phrases is also subject to linear proximity effects (Haskell & MacDonald 2005, Keung & Staub 2018, Peterson 1986). For example, it has been shown for disjoined subject phrases including number-mismatching conjuncts that the verb form that agrees with the linearly closer conjunct (e.g., *Can you tell me whether the horses or the clock is/are red?*) is selected or produced more often or rated higher than those agreeing with the first conjunct by English speakers.

In this study we investigate the extent to which these cross-linguistic observations concerning verb agreement hold in Turkish. Specifically, we examine the availability of partial agreement on the Turkish verb with both coordinated and disjoined subject phrases. In the remainder of the paper, we review previous work on conjunct agreement phenomena in Turkish in Section 2, present the methodology and results of three acceptability judgment tasks in Sections 3-5 and discuss the findings and lay out future directions in Section 6.

2 Previous work on conjunct agreement in Turkish

In Turkish both the subject-verb and genitive-possessive relations exhibit agreement, bearing person and number features (Kornfilt 1997). While full agreement is the standard, partial agreement is reported to be an available strategy in both nominal and verbal domains. The canonical word order in Turkish is SOV.

Tat & Kornfilt (2022) present data, including those from corpora, to illustrate cases of partial agreement in nominal phrases and clauses in Turkish. They show that when the possessor noun in a DP or the embedded subject in a nominalized clause is a coordination phrase, the possessee noun (as in 3a) or embedded verb (as in 3b) may have full agreement (i.e., 1PL for 1&2SG; see Ince & Aydın 2015) or partial agreement with the last/second conjunct only (i.e., 1SG).

- (3) a. [sen-im ve ben-im] hayal-ler-imiz/im/*in
 you-GEN and I-GEN dream-PL-1PL/1SG/2SG.POSS
 ‘your and my dreams’
 b. [o-nun ve ben-im] oyuncuyu yakından
 s/he-GEN and I-GEN actor closely
 tanı-ma-mız/m/*sı
 know-NMLZ-1PL/1SG/3SG.POSS
 ‘her/his and my knowing the actor closely’

Partial agreement strategies have also been observed and experimentally attested for the verbal domain (Köylü 2018). Using a rating and a forced choice task, Köylü examined the acceptability of sentences with coordinated subjects, in which the verb either agreed with the sum of the features of both conjuncts or the first or the second conjunct only, i.e., 1PL, 3SG and 1SG in 4², respectively.

- (4) [O ve ben] git-ti-k/m/*∅.
 s/he and I go-PST-1PL/1SG/3SG
 ‘S/he and I went.’

The rating task revealed that sentences with full agreement received higher ratings than those with partial agreement, regardless of the subject’s linear position with respect to the verb, i.e., preverbal or noncanonical postverbal. Within partial agreement, sentences with agreement with the linearly closer conjunct (e.g., *ben* ‘I’ in 4) were rated as more acceptable than those with agreement with the linearly distant conjunct. This preference was further confirmed by a forced-choice task, in which Turkish speakers reliably chose sentences with partial agreement with linearly closer conjuncts over those with partial agreement linearly distant conjuncts.

Both Tat & Kornfilt (2022) and Köylü (2018) propose a two-stage process for full/partial agreement, involving a syntactic and a post-syntactic component. However, they differ with respect to the computation of ϕ features and how partial agreement is realized. For Köylü, the syntactic stage involves an Agree relation between the agreement head and a conjunction phrase that inherits the features of both conjuncts, leading to full agreement. Partial agreement with the linearly closer conjunct, however, is a PF phenomenon that happens after spell-out. Tat & Kornfilt also argue for a two-stage process. They propose that in the syntactic stage interpretable features of the goal are checked against uninterpretable features of the probe. The morphological realization of these features is a separate, post-syntactic process that happens after spell-out and before the phonological form is received. This morphological step is where the features can be altered during lexical insertion, leading to partial agreement.

Lastly, Paparounas & Akkuş (2024) noted cases of default agreement on the verbs of nominalized embedded clauses. They showed that for some Turkish speakers, when the subject of an argument nominalized clause is a coordination phrase including a local and a non-local person such as ‘I and Leyla’ in 5, the Agree probe

²The example in 4 is constructed based on the descriptions in Köylü’s 2018 paper.

of the genitive case *-(n)In* cannot be valued, resulting in the insertion of the default 3SG agreement on the verb (see Paparounas & Akkuş 2024 for other, non-coordinated DPs that lead to default agreement).

- (5) Herkes [ben ile Leyla]-nın beraber gel-diğ-in-i
 everyone I and Leyla-GEN together come-NMLZ-3SG.POSS-ACC
 gör-dü.
 see-PST
 ‘Everyone saw that I and Leyla came together.’

Overall, while there is evidence for partial agreement in Turkish, the contexts in which it is attested and empirically tested remain limited. This study expands on prior research by using three acceptability judgment tasks to investigate agreement with coordinated and disjuncted subject phrases in the Turkish verbal domain. To anticipate the results, we find that Turkish speakers have an overall preference for full agreement resolving the features of both conjuncts. This preference is less pronounced with disjuncted subjects, which are generally less acceptable than coordinated subjects. There is also evidence for acceptability of partial agreement with the second conjunct for both coordinated and disjuncted subject phrases combining a proper name and a local pronoun (e.g., *Yıldız and I*).

3 Experiment 1: NP vs. clausal coordination account

Experiment 1 was designed to determine if partial agreement is a genuine agreement strategy in Turkish, or if it is a byproduct of other syntactic processes involving regular agreement strategies. It has been argued that partial agreement could result from an underlying structure in which constituents larger than just NPs, such as clauses, are coordinated (Aoun *et al.* 1994). This structure would then undergo Right Node Raising and/or ellipsis, which makes it appear as if the verb only agrees with one of the conjuncts. This is illustrated with a simplified configuration in 6.

- (6) [[IP NP₁ ~~Verb~~] and [IP NP₂ Verb]]

Previous experimental work on Turkish provided evidence for partial agreement with the linearly closer conjunct in Turkish (Köylü 2018), but all the verbs in the stimuli were distributive verbs that allow for a clausal coordination reading. In Experiment 1, we investigate whether partial agreement is also possible with collective verbs such as *gather* which do not allow for clausal coordination. This experiment was preregistered at: <https://osf.io/kxb6a/overview>

3.1 Participants

69 native speakers of Turkish ($M_{age} = 24.5$) participated in the task. Participants (including those in Experiments 2 & 3) were recruited through social media. Each participant was compensated 100 Turkish Lira (~\$3.5 at the time of the study).

3.2 Materials

The study had a 2x3 design, crossing PREDICATE (i.e., distributive such as *telaşlan-* ‘to worry’ or collective such as *toplan-* ‘to gather’), and AGREEMENT on the verb (i.e., Full Agreement (FA) bearing 1PL, Last Conjunct Agreement (LCA) bearing 1SG or First Conjunct Agreement (FCA) bearing the morphophonologically null 3SG). All stimuli had preverbal subjects including a proper name followed by the 1SG pronoun coordinated with *ve* ‘and’ as in *Yıldız ve ben* ‘Yıldız and I’. There was an adverbial phrase before the verb to reveal whether partial agreement would arise when the subject was not adjacent to the verb. An example set of stimuli is in 7.

- (7) [Yıldız ve ben] ofisteki görüşme için telaşlan-/toplan-dı-k/m/Ø.
Yıldız and I at.office meeting for worry-/gather-PST-1PL/1SG/3SG
‘Yıldız and I {worried about, gathered for} the meeting at the office.’

Six experimental lists were created in a Latin-square design. Each list included 36 target items and 72 fillers. 24 of the fillers were grammatical, 24 of them were ungrammatical, and the last 24 had coordinated objects.

3.3 Procedure

Experiment 1 (and Experiments 2 & 3) was conducted online on PCIBex (Zehr & Schwarz 2018). In the task, each sentence was presented as a whole, and the participants were instructed to rate them on a scale of 1-7, where 7 indicated ‘definitely a sentence that I might produce or hear from Turkish speakers’ and 1 ‘definitely not a sentence that I might produce or hear from Turkish speakers’. With instructions emphasizing daily speech, we aimed to tap into the acceptability of sentences in daily speech rather than their acceptability in written forms per prescriptive rules. Each sentence was followed by a comprehension question. There were three practice trials, for which feedback was provided.

3.4 Data analyses

The data were analyzed with mixed effects models (Bates *et al.* 2015) using the CLMM function in the `ordinal` package in R (Christensen 2018). The dependent variable was acceptability ratings. Fixed effects were PREDICATE and AGREEMENT, with participants and items as random effects. PREDICATE was coded using sum contrasts. AGREEMENT was coded using Helmert contrasts, which compared the effects of FULL (FA vs. LCA & FCA) and LAST CONJUNCT AGREEMENT (LCA vs. FCA). We fit two models: a crossed and a nested model. The crossed model had both fixed effects and their interaction. The nested model had a main effect of PREDICATE and separate AGREEMENT contrasts within distributive and collective predicate conditions. Each model (in Experiments 1-3) included random intercepts for participants and items. Absolute *z*-scores that are 2 or above indicate significance at the α level .05.

3.5 Results

Before the main analyses, we first inspected participants’ accuracy in the filler items. All participants had a mean accuracy above 70%; so they were all included in the analyses. The distribution of participants’ ratings is given in Figure 1.

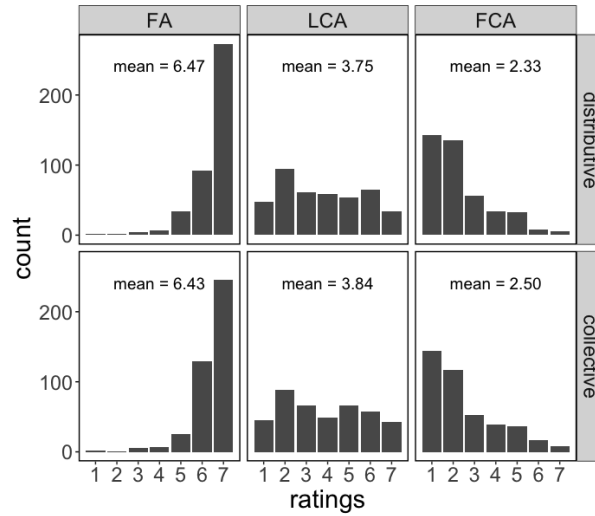


Figure 1: Distribution of ratings (7 = ‘definitely acceptable’) in Experiment 1, showing the count of responses per rating and mean ratings per condition. FA, LCA and FCA stand for Full, Last Conjoint and First Conjoint Agreement, respectively.

The results of the statistical analyses³ showed that overall, sentences with FA were more acceptable than those with partial agreement, i.e., LCA or FCA ($\hat{\beta} = 5.01$, $SE = 0.14$, $z = 37.08$). Within partial agreement strategies, sentences with LCA were more acceptable than those with FCA ($\hat{\beta} = 1.74$, $SE = 0.10$, $z = 18.35$), leading to the FA > LCA > FCA pattern in acceptability. There was an interaction between PREDICATE and FULL AGREEMENT ($\hat{\beta} = 0.41$, $SE = 0.18$, $z = 2.30$). Pairwise comparisons revealed the nature of this interaction: While the FA > LCA > FCA pattern holds for both distributive and collective verbs, the differences were slightly larger for distributive verbs ($\hat{\beta}$ ’s ≥ 1.72 , $SE \geq 0.13$, $z \geq 13.19$). There were no effects of PREDICATE or an interaction between PREDICATE and LAST CONJUNCT AGREEMENT (z ’s ≤ 0.22).

To further investigate the availability of FCA as an agreement strategy, we compared the ratings for FCA sentences and ungrammatical fillers. A paired-samples *t*-test showed no evidence for a difference between FCA ($M = 2.41$, $SE = 0.15$) and ungrammatical sentences ($M = 2.43$, $SE = 0.09$), $t(68) = 0.16$, $p = .87$).

3.6 Discussion

Experiment 1 showed that overall, sentences with full agreement received the highest ratings. Within the two partial agreement strategies, sentences with agreement with the linearly closer conjunct received higher ratings than those with agreement with the linearly distant conjunct, yielding the FA > LCA > FCA pattern. Crucially, this was the case with both distributive and collective verbs, with even less nuanced differences and slightly better ratings for partial agreement with collective verbs. These findings suggest that the availability of LCA cannot be merely due

³Full output from the statistical models for Experiments 1-3 is available at: <https://osf.io/c2pqh/files>

to a clausal coordination reading, but rather it is available as an agreement strategy with cases of NP-coordination. A limitation of Experiment 1 is that the low ratings observed for FCA may have been influenced by the morphophonologically null 3SG morphological marking. We address this potential confound in Experiment 2.

4 Experiment 2: The role of conjunct type and conjunct order

Experiment 1 demonstrated evidence for partial agreement with the second conjunct in coordinated subject phrases structured as [Proper Name + 1SG pronoun] (e.g., [Yıldız and I]) that were in a canonical, preverbal position. The availability of partial agreement with the highly prominent 1SG pronoun may stem from either its linear proximity to the verb or its privileged status as the 1SG pronoun. To disentangle these two factors, this experiment directly compares partial agreement strategies across two types of coordinated subject phrases: [Pronoun + Pronoun] (composed of 1SG and 2SG pronouns) and [Name + Pronoun] (composed of a proper name and a 1SG pronoun). This comparison will reveal whether partial agreement with ‘I’ is dependent on it being the only pronoun. Furthermore, the experiment manipulates the order of the coordinated subject phrase by placing it either preverbally or postverbally (i.e., noncanonically), and systematically positions the 1SG conjunct as the linearly distant one from the verb. This aims to reveal whether linear proximity with the verb is necessary condition for partial agreement. This experiment was preregistered at: <https://osf.io/kvw8f/overview>

4.1 Participants

94 native speakers of Turkish ($M_{age} = 25.2$) participated in the task. Each participant were compensated 120 Turkish Lira (~\$3.5 at the time of the study).

4.2 Materials

The study had a 2x2x3 design, crossing ORDER (i.e., preverbal or postverbal subjects), CONJUNCT (i.e., coordinated subjects combining the 1SG and 2SG pronouns or the 1SG pronoun and a proper name), and AGREEMENT on the verb as in the previous experiment (i.e., FA, LCA with the second conjunct or FCA with the first conjunct). All the stimuli had subjects coordinated with *ve* ‘and’. Differently from Experiment 1, the preverbal subject conditions had the 1SG pronoun as the initial conjunct in the subject phrase, making it the conjunct that is linearly more distant from the verb. To keep the linear distance between the 1SG pronoun and the verb constant across conditions, the order of the conjuncts was reversed in postverbal subjects, placing the 1SG pronoun as the second/last conjunct. Unlike Experiment 1, Experiment 2 tested FCA sentences featuring an overt morphological marking to establish whether the availability of FCA depends on the overt realization of agreement. An example set of stimuli is in 8.

- (8) a. [Ben ve Yıldız/sen] uzun zaman bekle-di-k/n/Ø/m.
 I and Yıldız/you long time wait-PST-1PL/2SG/3SG/1SG
 ‘I and {Yıldız, you} waited for a long time.’

- b. Bekle-di-k/m/n/∅ uzun zaman [Yıldız/sen ve ben].
 wait-PST-1PL/1SG/2SG/3SG long time Yıldız/you and I.
 ‘{Yıldız, you} and I waited for a long time.’

Twelve experimental lists were created in a Latin-square design, each containing 48 target items and the same 72 filler items from Experiment 1.

4.3 Data analyses

Fixed effects were ORDER, CONJUNCT and AGREEMENT; and random effects were participants and items. We ran a crossed model and a nested model. The crossed model included all three fixed effects and their two- and three-way interactions. The nested model had main effects of ORDER and CONJUNCT and separate AGREEMENT contrasts within each unique combination of CONJUNCT and ORDER, i.e., preverbal subjects with two pronouns, preverbal subjects with a proper name and the same for postverbal subjects. The investigation of agreement contrasts within each such combination allowed us to closely examine how the acceptability of agreement patterns changed in relation to conjunct properties such as order and pronominal status. ORDER and CONJUNCT were sum-contrast coded, and AGREEMENT was coded using Helmert contrast as in Experiment 1, in order to investigate FULL and LAST CONJUNCT AGREEMENT. The rest of the analysis steps was the same as before.

4.4 Results

Four participants’ accuracy was below 70% in the filler items; hence, they were excluded from the analyses. The mean accuracy of the remaining 90 participants was 94.4%. The distribution of participants’ ratings is given in Figure 2.

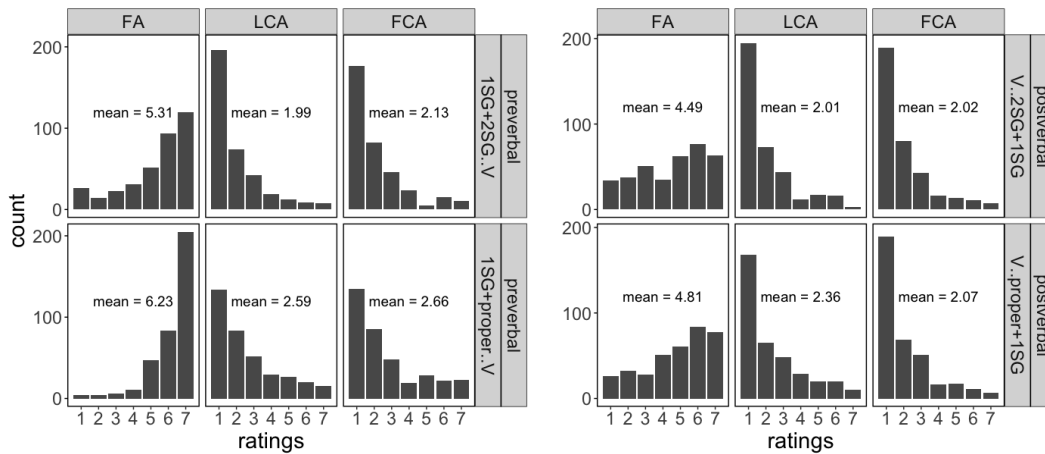


Figure 2: Distribution of ratings (7 = ‘definitely acceptable’) in Experiment 2, showing the count of responses per rating and mean ratings per condition. Preverbal conjuncts are on the left; postverbal conjuncts are on the right. FA, LCA and FCA stand for Full, Last Conjunct and First Conjunct Agreement, respectively.

The results of the statistical analyses showed that overall, sentences with a proper name and those with preverbal subjects were more acceptable than those

with two pronouns and those with postverbal subjects, respectively ($\hat{\beta}$'s ≥ 0.68 , SE 's ≥ 0.06 , z 's ≥ 11.36). For agreement, sentences with FA were more acceptable than those with partial agreement ($\hat{\beta} = 4.10$, $SE = 0.09$, $z = 46.06$); within partial agreement, there was no evidence for an overall difference between sentences with LCA and FCA ($\hat{\beta} = 0.01$, $SE = 0.08$, $z = 0.11$). All two-way interactions between the predictors were significant ($\hat{\beta}$'s ≥ 0.29 , SE 's ≥ 0.15 , z 's ≥ 2.01), but since our focus is on agreement patterns within unique subsets of data combining different conjunct types and orders, we do not examine those interactions further.

There was also a trend for a three-way interaction between ORDER, CONJUNCT and FULL AGREEMENT ($\hat{\beta} = 0.50$, $SE = 0.26$, $z = 1.93$), but it did not reach significance. The comparisons of agreement patterns within combinations of different conjunct types and order revealed that sentences with FA were more acceptable than those with partial agreement with all conjunct types in either order ($\hat{\beta}$'s ≥ 3.42 , SE 's ≥ 0.14 , z 's ≥ 25.34). When the subject was postverbal and included a proper name, sentences with LCA were more acceptable than those with FCA ($\hat{\beta} = 0.40$, $SE = 0.14$, $z = 2.75$). There were no other effects (z 's ≤ 1.74).

4.5 Discussion

Confirming Experiment 1, the findings of Experiment 2 showed a robust preference for full agreement for coordinated subject phrases, across all tested conjuncts.

Furthermore, Experiment 2 revealed an additional partial agreement strategy. For coordination phrases composed of the 1SG and 2SG pronouns, both FCA and LCA received similarly low ratings. Conversely, when the subject phrases included a proper name followed by the 1SG pronoun in the second conjunct position, there was evidence for partial agreement with the pronoun but it was limited to the postverbal environment. Given the surface order in this configuration (i.e., *V [Yıldız and I]*), the 1SG pronoun was the linearly distant element from the verb. Crucially, there was no evidence for LCA with the pronoun when it was the first conjunct in a preverbal subject phrase (i.e., *[I and Yıldız] V*). Taken together with the findings from Experiment 1, these results suggest that LCA arises when the 1SG pronoun was the second conjunct in a coordinated subject phrase combined with a proper name, regardless of its proximity to the verb in the surface linear order.

5 Experiment 3: Agreement with disjunction phrases

The goal of Experiment 3 is to reveal the verbal agreement strategies with disjunction phrases in subject position including *veya* 'or' in Turkish, and to what extent these strategies are similar to or different from those with coordination phrases.

5.1 Participants

93 native speakers of Turkish ($M_{age} = 23.5$) participated in the task. Each participant was compensated 120 Turkish Lira (~\$3.5 at the time of the study).

5.2 Materials

The study had a 2x2x3 design, crossing CONJUNCTION (i.e., coordination with *ve* ‘and’ or disjunction with *veya* ‘or’), PREDICATE (i.e., verbal such as *gel-* ‘to come’ or nonverbal such as *hazır-* ‘to be ready’), and AGREEMENT on the verb as in the previous experiments (i.e., Full Agreement (FA) with 1PL, Last Conjunct Agreement (LCA) with 1SG or First Conjunct Agreement (FCA) with 3SG). All the stimuli had preverbal subjects including a proper name followed by the 1SG pronoun, as in Experiment 1. An example set of stimuli is in 9.

- (9) [Yıldız ve/veya ben] kahve için gel-/hazır-dİ-k/m/Ø.
Yıldız and/or I coffee for come-/be.ready-PST-1PL/1SG/3SG
‘Yıldız {and, or} I {came, were ready} for the coffee.’

Twelve experimental lists were created in a Latin-square design, each containing 48 target items and the same 72 filler items from Experiment 1.

5.3 Data analyses

Fixed effects were CONJUNCTION, PREDICATE and AGREEMENT; and random effects were participants and items. Since PREDICATE did not modulate agreement patterns, we aggregated ratings for verbal and nonverbal predicates, and conducted analyses using the COORDINATION and AGREEMENT predictors only (see below for details). We ran a crossed model and a nested model. The crossed model included CONJUNCTION and AGREEMENT as fixed effects and their interaction. The nested model had a main effect of CONJUNCTION and separate AGREEMENT contrasts within coordination and disjunction conditions. CONJUNCTION was sum-contrast coded, and AGREEMENT was coded using Helmert contrast as in Experiments 1 and 2, in order to investigate FULL and LAST CONJUNCT AGREEMENT. The rest of the analyses was the same as before.

5.4 Results

Two participants’ accuracy was below 70% in the fillers; so, they were excluded from the analyses. The mean accuracy of the remaining 91 participants was 96.3%.

The most complex model, which included all three predictors, i.e., COORDINATION, PREDICATE, and AGREEMENT, showed no evidence for a three-way interaction between PREDICATE and the other two predictors (z ’s ≤ 0.71). Given that PREDICATE type (i.e., verbal vs. nonverbal) did not modulate the critical differences for AGREEMENT, the ratings for verbal and nonverbal predicates were aggregated for subsequent analyses. The distribution of participants’ ratings after this aggregation is shown in Figure 3.

The results of the analyses revealed that overall, sentences with coordination were more acceptable than those with disjunction ($\hat{\beta}$ ’s ≥ 1.80 , SE ’s ≥ 0.06 , z ’s ≥ 28.11). For agreement, sentences with FA were more acceptable than those with partial agreement ($\hat{\beta} = 3.26$, $SE = 0.08$, $z = 40.74$); and within partial agreement, sentences with LCA were more acceptable than those with FCA ($\hat{\beta} = 1.60$, $SE = 0.07$, $z = 22.40$), replicating the FA > LCA > FCA pattern. There were also interactions between CONJUNCTION and FULL AGREEMENT, and CONJUNCTION and

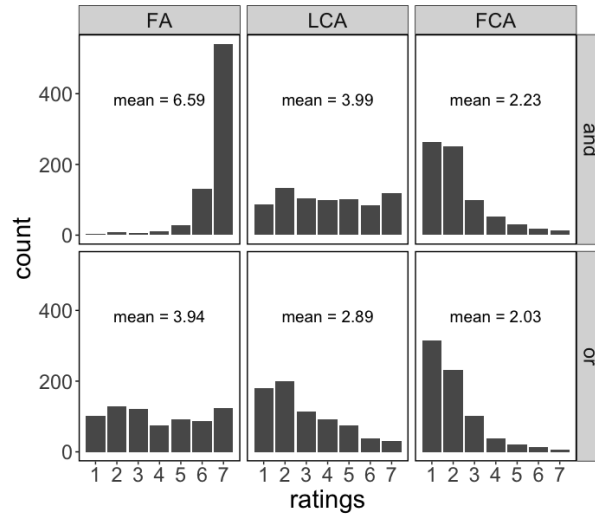


Figure 3: Distribution of ratings (7 = ‘definitely acceptable’) in Experiment 3, showing the count of responses per rating and mean ratings per condition. The ratings for PREDICATE were aggregated, in line with the analyses. FA, LCA and FCA stand for Full, Last Conjoint and First Conjoint Agreement, respectively.

LAST CONJUNCT AGREEMENT ($\hat{\beta}$ ’s ≥ 0.93 , SE ’s ≥ 0.14 , z ’s ≥ 6.86). While the FA > LCA > FCA pattern holds for sentences with both coordination and disjunction, the differences were larger for coordination ($\hat{\beta}$ ’s ≥ 1.13 , SE ’s ≥ 0.10 , z ’s ≥ 11.62).

5.5 Discussion

Experiment 3 provided a direct comparison for verbal agreement strategies with coordination and disjunction phrases in preverbal subject position. The results revealed that compared to sentences with coordination phrases with *ve* ‘and’, the ratings for sentences with disjunction phrases with *veya* ‘or’ were overall degraded. This was the case with both full and partial agreement. This provides novel evidence from Turkish supporting the hypothesis that no single form of agreement is fully acceptable with disjunction. Despite the low ratings, the FA > LCA > FCA pattern observed with coordination extends to disjunction, indicating that agreement with disjunction phrases is also subject to linearity effects, though the statistical differences among agreement options were smaller than those for conjunction.

6 General Discussion

The present study investigated verbal agreement strategies in Turkish for subjects involving coordination or disjunction phrases via three acceptability judgment tasks. For coordination phrases, there was a strong preference for full agreement, where the verb exhibits the sum of the features of both conjuncts (i.e., 1PL for 1&2/3SG). Partial agreement with the 1SG pronoun was also *somewhat* available, with both distributive and collective verbs. Crucially, partial agreement was limited to cases when the 1SG pronoun was the second conjunct (e.g., *Yıldız and I*), and this was so for both preverbal and postverbal subjects. Partial agreement was unavailable if

this pronoun appeared as the first conjunct (e.g., *I and Yıldız*) or if the coordinated phrase contained 1SG and 2SG pronouns together (e.g., *you and I*). For disjunction phrases (investigated only in preverbal position), no agreement strategy was rated as fully acceptable. Nevertheless, full agreement was again the most preferred option, with limited availability for partial agreement with the 1SG pronoun as the second conjunct in disjunction with a proper name (e.g., *Yıldız or I*).

Our goal in Experiment 1 was to determine whether partial agreement in Turkish functions as a genuine agreement strategy or if it is merely a result of a clausal coordination analysis (Aoun *et al.* 1994), where the verb agrees with a single conjunct in its clause, but surfaces as the verb form for the whole subject phrase after operations such as ellipsis or Right Node Raising. To test this, we used collective verbs, which enforce the coordination of NPs, thereby eliminating the clausal coordination structure. Our results showed that partial agreement with the second conjunct is available with both collective and distributive verbs. The statistical results also showed that the differences in acceptability between full and partial agreement were less pronounced for collective than distributive verbs. This finding was particularly interesting as it is the opposite of what would be predicted by the clausal coordination explanation. The results thus provide strong evidence against clausal coordination analysis being the primary source of partial agreement, but rather support that it is a genuinely available agreement strategy in Turkish (see Arsenijević *et al.* 2020 for similar empirical findings in South Slavic languages).

For partial agreement with coordination, Experiments 1 and 2 revealed that among the conjunct combinations that we tested, the only one that enabled partial agreement was a proper name followed by the 1SG pronoun (i.e., ‘*[Yıldız and I] V*’ or ‘*V [Yıldız and I]*’) (cf. Köylü 2018). Crucially, this pattern held for both preverbal and postverbal subjects. While the 1SG pronoun is the linearly closer conjunct with respect to the verb when the subject phrase is preverbal, it is the linearly distant conjunct in postverbal subjects. This outcome cannot be easily accounted for by the hierarchical position of the conjuncts as the second conjunct is the hierarchically lower one. It is also difficult to reconcile this finding with linearly-sensitive partial agreement accounts (e.g., Bhatt & Walkow 2013, Tat & Kornfilt 2022), which attribute partial conjunct agreement to proximity between the agreeing conjunct and the verb in the surface linear order. Due to the limited range of combinations currently tested, we cannot offer a conclusive analysis here; we instead discuss potential explanations and consider further directions.

At first glance, the results above may seem to suggest that partial agreement is determined by the *linearly second position* of the agreeing conjunct within the coordination phrase. This conclusion arises because the verb can access the 1SG pronoun when it is the second conjunct, regardless of the subject phrase’s position relative to the verb in the surface linear order (e.g., *[Yıldız and I] V* and *V [Yıldız and I]*), but not when the 1SG pronoun is the first conjunct in a preverbal subject phrase (e.g., *[I and Yıldız] V*).

Given current theories, it seems unlikely that partial agreement is solely conditioned on a conjunct’s linear position internal to the subject phrase. Rather, we consider it more plausible that Agree relations are established based on the conjunct’s linear proximity to the verb in the canonical word order (where the second conjunct would typically be closer to the verb), and that scrambling, which yields

non-canonical word orders like the postverbal subject cases here, applies after the Agree and initial linearization steps. However, we refrain from drawing a conclusion at the moment given the limitations of our current data. To disentangle the roles of the conjunct's position within the phrase and its linear proximity to the verb, future investigations must examine coordinated subjects in which the pronoun is the first but also the linearly closer conjunct to the verb (e.g., *V [I and Yıldız]*).

Moreover, to determine whether the observed partial agreement extends beyond the 1SG pronoun or if it arises specifically due to the 1SG's highly prominent person features, future studies should examine coordination phrases excluding the 1SG pronoun (e.g., *Yıldız and you/she*). This would also help clarify the lack of partial agreement in the current stimuli that included both 1SG and 2SG pronouns, indicating whether this is due to a feature competition or a separate structural constraint.

For disjunction phrases, our results from Experiment 3 provided novel evidence for the lack of a fully acceptable agreement form from Turkish, consistent with findings from English and Italian (Foppolo & Staub 2020, Keung & Staub 2018, Haskell & MacDonald 2005). We interpret the low acceptability with disjunction phrases as supporting the hypothesis that a grammatical gap exists when assigning verb number to disjuncted subjects (Foppolo & Staub 2020). This gap may force speakers to rely on ad hoc, probabilistic strategies for agreement. The reliance on such non-deterministic strategies may introduce a processing cost due to metalinguistic awareness, leading to the observed degraded ratings.

Despite the overall degraded ratings, Turkish speakers exhibited a preference for full agreement over partial agreement for disjuncted phrases, consistent with Italian production data (Foppolo & Staub 2020), though this preference was less pronounced than the one with coordination phrases. Partial agreement with the 1SG pronoun, i.e., the linearly closer conjunct, (e.g., [*Yıldız or I*] *V*) was the next most acceptable strategy. These findings demonstrate that the FA > LCA > FCA pattern established for preverbal coordination phrases extends to disjunction phrases in Turkish. However, we cannot conclude that all partial agreement constraints and patterns for coordination extend to disjunction, as Experiment 3 tested only a limited combination and only in the preverbal position. To fully map the similarities and differences in agreement strategies, future work must systematically test disjuncted subjects in the postverbal position and with a wider variety of conjuncts.

While the current study establishes the robustness of partial agreement as a genuinely available strategy in Turkish, further work is necessary to fully map the range of its acceptability and constraints. It is possible that the availability of partial agreement varies across and/or within speakers of Turkish. For instance, Marušič *et al.*'s 2015 experimental work on gender agreement with coordinated subjects in Slovenian documented notable inter- and intra-speaker variation, suggesting the co-existence of multiple agreement grammars. To determine if similar variability also exists for Turkish speakers, more fine-grained statistical methods, such as Bayesian analyses that quantify individual variation, should be applied to the current dataset. We leave this to future work. Lastly, the overarching goal of this project is to formulate a comprehensive conjunct agreement system for Turkish. While the work presented here is a crucial step toward that goal, more systematic investigation should examine crucial combinations of conjuncts varying in person, pronominal status, and linear order, to reveal the full extent of conjunct agreement.

References

- Aoun, J., E. Benmamoun, & D. Sportiche. 1994. Agreement, word order, and conjunction in some varieties of Arabic. *Linguistic Inquiry* 25. 195–220.
- Arsenijević, B., J. Willer-Gold, N. Aljović, N. Čordalića, M. Kresić Vukosav, N. Leko, F. Malenica, F. L. Marušić, T. Milićev, N. Milićević, *et al.*. 2020. Elided clausal conjunction is not the only source of closest-conjunct agreement: A picture-matching study. *Syntax* 23. 78–104.
- Bates, D., M. Maechler, B. Bolker, & S. Walker. 2015. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67. 1–48.
- Benmamoun, E., A. Bhatia, & M. Polinsky. 2009. Closest conjunct agreement in head final languages. *Linguistic Variation Yearbook* 9. 67–88.
- Bhatt, R., & M. Walkow. 2013. Locating agreement in grammar: An argument from agreement in conjunctions. *Natural Language & Linguistic Theory* 31. 951–1013.
- Bošković, Ž. 2009. Unifying first and last conjunct agreement. *Natural Language & Linguistic Theory* 27. 455–496.
- Christensen, R. H. B. 2018. Cumulative link models for ordinal regression with the R package ordinal.
- Citko, B. 2018. Complementizer agreement with coordinated subjects in Polish. *Glossa: a journal of general linguistics* 3. 124.
- Foppolo, F., & A. Staub. 2020. The puzzle of number agreement with disjunction. *Cognition* 198. 104–161.
- Haskell, T. R., & M. C. MacDonald. 2005. Constituent structure and linear order in language production: Evidence from subject-verb agreement. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 31. 891–904.
- Ince, A., & Ö. Aydın. 2015. Copular structures as (non)phases: Synchronic and diachronic perspective. In *Turcologica*, ed. by D. Zeyrek, Sağın Şimşek, J. Rehbein, & U. Atas, volume 113 of *Ankara Papers in Turkish and Turkic Linguistics*, 87–97. Wiesbaden: Harrassowitz Verlag.
- Keung, L.-C., & A. Staub. 2018. Variable agreement with coordinate subjects is not a form of agreement attraction. *Journal of Memory and Language* 103. 1–18.
- Kornfilt, J. 1997. *Turkish*. London: Routledge.
- Koutsoudas, A. 1968. The A over A convention. *Linguistics* 46. 11–20.
- Köylü, Y. 2018. A compositional approach to conjunct agreement in Turkish. *Proceedings of the Linguistic Society of America* 3. 17:1–12.
- Marušić, F., A. I. Nevins, & W. Badecker. 2015. The grammars of conjunction agreement in Slovenian. *Syntax* 18. 39–77.
- Munn, A. 1993. *Topics in the syntax and semantics of coordinate structures*. College Park: University of Maryland dissertation.
- Murphy, A., & Z. Puškar. 2018. Closest conjunct agreement is an illusion. *Natural Language & Linguistic Theory* 36. 1207–1261.
- Nevins, A., & P. Weissner. 2019. Closest conjunct agreement. *Annual Review of Linguistics* 5. 219–241.
- Paparonas, L., & F. Akkuş. 2024. Anaphora and agreement in the Turkish DP: Delimiting binding-through-Agree. *Natural Language & Linguistic Theory* 42. 633–700.
- Peterson, P. G. 1986. Establishing verb agreement with disjunctively conjoined subjects: Strategies vs. principles. *Australian Journal of Linguistics* 6. 231–249.
- Peterson, P. G. 2004. Coordination: Consequences of a lexical-functional account. *Natural Language & Linguistic Theory* 22. 643–679.
- Tat, D., & J. Kornfilt. 2022. Partial versus full agreement in Turkish possessive and clausal DP coordination. In *Pseudo-Coordination and Multiple Agreement Constructions*, ed. by G. Giusti, V. N. Di Caro, & D. Ross, 271–286. Amsterdam: John Benjamins.
- van Koppen, M. 2005. *One probe-two goals: Aspects of agreement in Dutch dialects*. Leiden: Leiden University dissertation.
- Zehr, J., & F. Schwarz. 2018. PennController for Internet Based Experiments (IBEX).